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Recommendations on calculation of transport coefficients and thermodynamic properties of helium-xenon gas mixtures☆

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A B S T R A C T

This paper discusses the results of a review of computational and experimental studies on the transport coefficients and thermodynamic properties of helium-xenon gas mixtures. The most promising in the authors' opinion is a combination of approaches pertaining to the kinetic Chapman-Enskog theory and the theory of corresponding states of dense gas.

This integrated approach yielded semi-empirical correlations for calculation of density, dynamic viscosity coefficient, thermal conductivity coefficient, specific thermal capacity at constant pressure and constant volume. These correlations account for the influence of temperature and pressure as well as for the proportion of mixture components. However they appear in a form unsuitable as yet for practical calculation of transport coefficients and thermodynamic properties of helium-xenon mixtures. In other words, it is still necessary to develop computation algorithms with indication of the calculation sequences, explanation/interpretation of the expressions for the first and second derivatives of the associated model parameters, etc.

The authors of the present paper developed such algorithms on a model of corresponding states of dense gas (CSDG), proceeding from the above semi-empirical relations. These algorithms are compatible with the simplified procedure which disregards the mixture composition effect and makes allowance for the influence of temperature and pressure by means of simple polynomial formulae.

1. Introduction

Helium-xenon mixtures are of potential interest for development of modern power equipment, and to do their job, the designers need to have as accurate information as possible concerning the density, viscosity, heat capacity and thermal conductivity of the candidate mixture.

However, the thermodynamic properties and transport coefficients of binary gas mixtures, with the helium-xenon mixture among them, were found to be non-linear functions of mixture concentration. A clear example of this is the Prandtl number which varies from 0.6 to 0.7 for pure components of the mixture and may be as low as 0.1 ÷ 0.4 when the components are mixed. On the other hand, it is well known that the Prandtl number affects such a vital parameter of power equipment as the coefficient of convective heat transfer to single-phase gas or liquid flows.

As shown by analysis of relevant literature, it would be too early to give up studies on thermodynamic properties and transport coefficients of binary gas mixtures. The authors of this paper venture to make practical recommendations for calculating such parameters on the strength of the examined publications.

2. Availability of information on thermodynamic properties and transport coefficients of helium-xenon mixtures

The thermodynamic properties and coefficients of transport of helium, xenon and their mixtures were studied in many experiments and calculations (Ratnikov and Tetel'baum, 1978; Bretshnaider, 1966; Haire and Vargo, 2007; Tournier and El-Glenk, 2008; Hashimoto et al., 1992; El-Glenk and Tournier, 2008; Mashtovsii, 1977; Johnson, 2006; Karevskii, 2010; Soldatov, 2010; Vargaftik, 1972; El-Glenk and Tournier, 2007; Thornton, 1960; Fastovskii et al., 1972; Kirov et al., 1974; Brokaw, 1974; Nedostup, 1990; Cederberg et al., 1969; Teplofizicheskie, 2007; Shashkov, 1973; Cederberg, 1963; Termodinamika, 1991; Tekhnicheskaya, 1981; Tekhnicheskaya, 1971; Rabinovich et al., 1976; Zubarev et al., 1989; Rabinovich et al., 1982; Stepchikov, 1974; Brokaw, 1961; Vanco, 1965; Huber et al., 1996; Dragunov et al., 2012). However, those efforts dealt mostly with pure helium or xenon, with much fewer investigations devoted to helium-xenon mixtures. Moreover, the experimental studies on He-Xe mixtures were staged either in a wide range of temperatures but at very low

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Nomenclature

Acronyms and designations

SPTP	Simplified Prediction of Thermophysical Properties
CSDG	corresponding states of dense gas
NIST	National Institute of Standards and Technology (USA)
B, B_{ii}	second virial coefficient in the equation of state of pure gas (mole/m^3), $B_{ii} = B$
B_{ij}	second virial coefficient in the equation of state of a mixture, accounting for interaction between unlike-molecules of a mixture, (m^3/mole), $i \neq j$
B_r	reduced second virial coefficient, $B_r = B/V^*$
$\bar{B}$	second virial coefficient for a gas mixture (m^3/mole)
B_{12}^*	collision integral function, $B_{12}^* = [5\Omega^{(1,2)*} - 4\Omega^{(1,3)*}]/\Omega^{(1,3)*}$
C, C_{iiii}	third virial coefficient for pure gas (mole^2/m^6), $C_{iii} = C_i$
C_{ij}	third virial coefficient accounting for interaction between unlike-molecules of a mixture, (mole^2/m^6), $i \neq j$
C_r	reduced third virial coefficient, $C_r = V^{*2}$
$\bar{C}$	third virial coefficient for a gas mixure (m^6/mole^2)
C_p	molar specific heat capacity of pure gas or mixture at constant pressure (J/(mole K))
C_v	molar specific heat capacity of pure gas or mixture at constant volume (J/mole K)
g_i	mass fraction of the i-th component in a gas mixture
H	molar enthalpy (J/mole)
k	Boltzmann constant $k = 1.380662 \times 10^{-23}$ (J/K)
m	mass of one gas molecule, $m = M/N_A$ (kg)
M	molecular mass of gas (kg/mole)
$\bar{M}$	molecular mass of a mixture (kg/mole)
n	exponent in the viscosity calculation formula
N	number of components in a gas mixture
P	pressure (Pa)
R_g	universal gas constant, $R_g = 8.31441$ (J/(mole K))
R_{He}	gas constant for helium, $R_{He} = \frac{8314,4126}{4.0026} = 2077.253$ (J/(kg K))
R_{Xe}	gas constant for xenon, $R_{Xe} = \frac{8314,4126}{131.29} = 63.329$ (J/(kg K))
r_i	volume fraction of the i-th component in a gas mixture
S_i	component entropy, (J/(kg K))
T	temperature (K)
T_o	reference temperature for enthalpy (K)
T_{cr}, T_{ii}^*	critical temperature of pure gas (K), $T_{ii}^* = T_{cr}$
T_{ij}^*	characteristic temperature for including the interaction between unlike-molecules of a mixture (K), $i \neq j$
$\bar{T}^*$	"pseudocritical" temperature of a mixture (K)
T_μ	temperature in the viscosity calculation formula (K)
U	molar internal energy (J/mole)
V_{ii}^*	characteristic molar volume of pure gas (m^3/mole), $V_{ii}^* = V^* = R_g(T_{cr}/P_{cr})_i$
V_{ij}^*	characteristic molar volume for including the interaction between unlike-molecules of a mixture (m^3/mole), $i \neq j$
$\bar{V}^*$	characteristic molar volume of a gas mixture (m^3/mole)
Z	compressibility factor

Greek symbols

χ_i	molar fraction of the i-th component in a gas mixture
$\Delta\lambda$	pressure (density) contribution to heat conductivity coefficient (W/(m K)), $\Delta\lambda = \lambda(T, P) - \lambda^o(T)$
$\Delta\mu$	pressure (density) contribution to dynamic viscosity coefficient (Pa s), $\Delta\mu = \mu(T, P) - \mu^o(T)$
θ	reduced temperature, $\theta = T/T_{cr}$
λ	thermal conductivity coefficient (W/(m K))
$\bar{\lambda}^*$	"pseudocritical" thermal conductivity coefficient of a mixture (W/(m K))
μ	dynamic viscosity coefficient (Pa s)
$\bar{\mu}^*$	"pseudocritical" coefficient of dynamic viscosity (Pa s)
υ	molar volume of gas (m^3/mole)
ρ	density (kg/m^3)
$\bar{\rho}$	molar density (mole/m^3), $\bar{\rho} = \rho/M = 1/\upsilon$
ψ	dimensionless function of reduced density

Subscripts and superscripts

$\wedge$	molar property (per mole)
–	gas mixture property
*	reduced, critical or "pseudocritical" value
CE	Chapman-Enskog theory
cr	critical value
o	diluted gas; reference parameter value at pressure 0.1 MPa
λ	coefficients in thermal conductivity formulae
μ	coefficients in dynamic viscosity formulae

pressures or at more or less acceptable pressures but in the low temperature range.

An example of a successful literature search along these lines is study (Tournier and El-Glenk, 2008) which systematised the results of numerous computational and experimental investigations on the transport coefficients and thermodynamic properties of pure inert gases and their binary mixtures. The discussed experimental studies on helium, neon, argon, krypton, xenon and their binary mixtures covered a pressure range of 0.1 through 20 MPa and temperatures up to 1400 K, with their results added up to form an extensive database.

This database may serve as a source of constants required for semi-empirical correlations which would make acceptable allowance for the effect of pressure and temperature on thermodynamic properties and transport coefficients of inert gases and their binary mixtures in the parameter ranges of practical interest. Besides, this array of experimental data may be a convenient platform for assessing the accuracy of such correlations.

The authors of work (Tournier and El-Glenk, 2008) combined the approaches offered by the kinetic Chapman-Enskogs theory for diluted gases and by the theory of corresponding states to allow for the dependence of thermodynamic properties and transport equations on pressure. Such combination makes it possible to extend the models developed for diluted gases to dense gases.

3. Binary gas mixture density

The density of a binary mixture is obtained by solving the following virial equation of state:

$$P = (R_g T) \times (\rho Z) = R_g T [\rho + \bar{B}\rho^2 + \bar{C}\rho^3], \quad (1)$$

where compressibility factor Z is represented by a second-order polynomial in the molar density ρ.

The molecular mass of a binary mixture, $\bar{M}$, is determined from this equation as follows:

$$\bar{M} = \chi_1 M_1 + \chi_2 M_2 \quad (2)$$

Virial coefficients B and C relate to intermolecular potentials. The kinetic gas theory offers the following equations for their evaluation, obtained by means of statistical mechanics:

$$\bar{B} = \chi_1^2 B_{11} + 2\chi_1\chi_2 B_{12} + \chi_2^2 B_{22} \quad (3)$$

$$\bar{C} = \chi_1^3 C_{111} + 3\chi_1^2\chi_2 C_{112} + 3\chi_2^2\chi_1 C_{122} + \chi_2^3 C_{222} \quad (4)$$

Here, virile coefficientsB_{ii} and $Ciii//B_{11}$ and C_{11} belong to pure components of the gas mixture and are described by relations derived with the use of experimental data.

Virial equation of state (1) with semi-empirical expressions for its virial coefficients describes gas pressure by an explicit function of molar density and temperature. It can be easily inverted to determine the gas density. As pointed out in (Tournier and El-Glenk, 2008), calculations by this equation yield results in satisfactory agreement with experiment.

Eq. (1) can be brought to the following form:

$$\rho^3 + a_2\rho^2 + a_1\rho + a_o = 0 \quad (5)$$

where $a_o = -\frac{p}{\bar{C}R_gT}$; $a_1 = \frac{1}{\bar{C}}$; $a_2 = \frac{\bar{B}}{\bar{C}}$

Will all substitutions made as per (Dragunov et al., 2012), the inverted equation of state will explicitly describe molar gas – or gas mixture – density as a function of pressure:

$$\rho = \left\{\frac{1}{6}\left(\frac{\bar{B}}{\bar{C}^2} + \frac{3P}{\bar{C}R_gT}\right) - \frac{1}{27}\frac{\bar{B}^3}{\bar{C}^3} + \left[\left(\frac{1}{3}\frac{1}{\bar{C}} - \frac{1}{9}\frac{\bar{B}^2}{\bar{C}^2}\right)^3 + \left\langle\frac{1}{6}\left(\frac{\bar{B}}{\bar{C}^2} + \frac{3P}{\bar{C}R_gT}\right) - \frac{1}{27}\frac{\bar{B}}{\bar{C}^3}\right\rangle^2\right]^{1/2}\right\}^{1/3} + \left\{\frac{1}{6}\left(\frac{\bar{B}}{\bar{C}^2} + \frac{3P}{\bar{C}R_gT}\right) - \frac{1}{27}\frac{\bar{B}^3}{\bar{C}^3} - \left[\left(\frac{1}{3}\frac{1}{\bar{C}} - \frac{1}{9}\frac{\bar{B}^2}{\bar{C}^2}\right)^3 + \left\langle\frac{1}{6}\left(\frac{\bar{B}}{\bar{C}^2} + \frac{3P}{\bar{C}R_gT}\right) - \frac{1}{27}\frac{\bar{B}}{\bar{C}^3}\right\rangle^2\right]^{1/2}\right\}^{1/3} - \frac{1}{3}\frac{\bar{B}}{\bar{C}} \quad (6)$$

where $\bar{B}$ and $\bar{C}$ are defined by relations (3) and (4), respectively.

As demonstrated in (Tournier and El-Glenk, 2008), given pressure below 2.0 MPa and molecular mass M of less than 70 g/mole, Z deviations from unity will not exceed 1%, wherefore density can also be calculated by the ideal gas law. However, with higher pressure and temperature (P = 7.0 MPa and T = 400 K), Z variations can be as large as, e.g., $Z_{He} = 1.024$, $Z_{Xe} = 0.85$. Given the same pressure but a higher temperature (1200 K), the error in the density of all gas mixtures under consideration calculated by the ideal gas law will not be greater than 2%, and will be less than 1% if molecular mass M is under 20 g/mole.

4. Coefficient of dynamic viscosity of a dense binary mixture

The dynamic viscosity of a binary mixture is obtained using the two-parameter law of corresponding states:

$$\bar{\mu}(\mathrm{T,\ P}) = \bar{\mu}^{\mathrm{O}}(\mathrm{T}) + \left(1 - \frac{1}{2.3}\right) \times \bar{\mu}^* \times \psi_\mu \times \left(\frac{0.291 \times \bar{V}^*}{\bar{M}} \times \rho\right) \quad (7)$$

In the case of a helium-containing binary mixture, allowance must be made for the fact that the dynamic viscosity of helium is independent of pressure:

$$\Delta\bar{\mu}^* = \chi_1 \Delta\mu_{\mathrm{cr1}} + \chi_2 \Delta\mu_{\mathrm{cr2}} \approx \chi_1 \Delta\mu_{\mathrm{cr1}} \quad (8)$$

Index "1" refers to the heavier component of the mixture (e.g., to Xe) and index"2", to helium.

As a result, expression (7) takes the following form:

$$\bar{\mu}(\mathrm{T,\ P}) = \bar{\mu}^{\mathrm{O}}(\mathrm{T}) + \chi_1 \times \left(1 - \frac{1}{2.3}\right) \times \mu_1^* \times \psi_\mu \times \left(\frac{0.291 \times \bar{V}^*}{\bar{M}} \times \rho\right) \quad (9)$$

Expressions for calculation of $\bar{\mu}^{\mathrm{O}}$ (T), μ_1^*, ψ_μ and $\bar{V}^*$, produced by approximation of the available experimental data, are presented in (Tournier and El-Glenk, 2008; Dragunov et al., 2012)

As shown in (Tournier and El-Glenk, 2008), calculations by the derived formulae agree within ± 3% with the experimental data for a rarefied helium-xenon mixture (86 points with pressure under 0.1 MPa and temperature less than 1553 K) as well as with calculations by the kinetic Chapman-Enskog theory for rarefied gases.

5. Heat conductivity coefficient of a dense binary mixture

Application of the two-parameter law of corresponding states to binary gas mixtures results in the following expression:

$$\bar{\lambda}(T,\ P) = \bar{\lambda}^o(T) + \left(1 - \frac{1}{2.94}\right) \times \bar{\lambda}^* \times \psi_\kappa \times \left(\frac{0.291 \times \bar{V}^*}{\bar{M}} \times \rho\right) \quad (10)$$

Here, the "pseudocritical" heat conductivity of the mixture, $\bar{\lambda}^*$, is calculated as:

$$\bar{\lambda}^* = 0.304 \times 10^{-4} \times \frac{(\bar{\mathrm{T}}^*)^{0.277}}{(\bar{\mathrm{M}})^{0.465} \times (0.291 \times \bar{V}^*)^{0.415}} \quad (11)$$

while the "pseudocritical" temperature and volume, $\bar{T}^*$ and $\bar{V}^*$, of the binary mixture are determined by the van der Waals Mixing Rules:

$$\bar{V}^* = \sum_{i=1}^{N}\sum_{j=1}^{N} \chi_i\chi_j V_{ij}^*, \quad (12)$$

$$\bar{V}^*\bar{T}^* = \sum_{i=1}^{N} \chi_i V_{ii}^* T_{ii}^*, \quad (13)$$

$$\bar{V}^*\bar{T}^* = \sum_{i=1}^{N}\sum_{j=1}^{N} \chi_i\chi_j V_{ij}^* T_{ij}^*. \quad (14)$$

Expressions for calculation of $\bar{\lambda}^{\mathrm{O}}$ (T), ψ_λ and $\bar{V}^*$, obtained by approximation of the existing experimental data, are found in (Tournier and El-Glenk, 2008; Dragunov et al., 2012)

As noted in (Tournier and El-Glenk, 2008), the proposed approach offers agreement within ± 5% with the measured heat conductivities of rarefied binary mixtures (including, helium-xenon), it is consistent with the kinetic Chapman-Enskog theory, and can adequately describe the available experimental data on heat conductivity of He-Ar and Ne-Ar mixtures.

6. Enthalpy and specific heat capacity of inert gases and their mixtures

Thermodynamic properties, such enthalpy, specific heat capacity, and adiabatic index C_p/C_v, can be calculated from classical thermodynamic equations/relations.

Enthalpy is obtained by integration with respect to temperature T for constant pressure (rarefied gas) followed by integration with respect to density ρ at constant temperature:

$$H(T,\ P) = H^o(T_o) + C_P^o(T - T_o) + \rho R_g T\left[\left(B - T\frac{dB}{dT}\right) + \rho\left(C - \frac{T}{2}\frac{dC}{dT}\right)\right] \quad (15)$$

where $\hat{C}_P^o = (5R_g/2)$ stands for the heat capacity of ideal rarefied gas at constant pressure.

Specific heat capacity of dense gas at constant pressure is determined by the following relation:

$$C_P = \left[\frac{\partial H}{\partial T}\right]_P = C_P^o + \rho R_g\left[\left(B - T\frac{dB}{dT} - T^2\frac{d^2B}{d^2T}\right) + \rho\left(C - \frac{T^2}{2}\frac{d^2C}{dT^2}\right)\right] + R_g T\left[\left(B - T\frac{dB}{dT}\right) + \rho\left(2C - T\frac{dC}{dT}\right)\right] \times \left[\frac{\partial \rho}{\partial T}\right]_P, \quad (16)$$

where

$$\left[\frac{\partial \rho}{\partial T}\right]_P = -\frac{(\rho + B\rho^2 + C\rho^3)/T + \frac{dB}{dT}\rho^2 + \frac{dC}{dT}\rho^3}{1 + 2B\rho + 3C\rho^2} \quad (17)$$

Calculation of specific heat capacity of dense gas at constant volume relies on the expression:

$$C_\upsilon = \left[\frac{\partial U}{\partial T}\right]_{\hat{\rho}} = \left[\frac{\partial H}{\partial T}\right]_{\hat{\rho}} - \left[\frac{\partial}{\partial T}\left(\frac{P}{\rho}\right)\right]_{\hat{\rho}} = C_v^o - \rho R_g T\left[\left(2\frac{dB}{dT} + T\frac{d^2B}{dT^2}\right) + \rho\left(\frac{dC}{dT} + \frac{T}{2}\frac{d^2C}{dT^2}\right)\right], \quad (18)$$

where $C_v^o = (3R_g/2)$ is heat capacity of ideal gas at constant pressure.

Letters B and C stand for, respectively, the second and third virial coefficients in the? equation of state, which are described as found in (Tournier and El-Glenk, 2008; Dragunov et al., 2012).

The values of enthalpy, specific heat capacity and $\gamma = C_p/C_v$ calculated by these formulae are in satisfactory agreement with the values recommended by the NIST (National Institute of Standards and Technology, USA) for He, Ne, Ar, Kr and Xe in the pressure range of 0.1 to 20 MPa and at temperatures, respectively, up to 1500, 700, 700, 800 and 800 K. Given temperatures in excess of (T_{cr} + 50 K), the difference between the calculated and recommended enthalpy values is no greater than 1%, and is under 0.1% beyond the critical range.

7. Prandtl number

Publication (Tournier and El-Glenk, 2008) presents the Prandtl number for pure gases and their binary mixtures in the pressure range between 2.0 and 7.0 MPa. At low densities (with pressures no higher than 0.1 MPa), the Prandtl number is independent of temperature and molecular mass and equals 0.667. At higher pressures, the Prandtl number of pure inert gas grows with the increase of gas pressure and molecular mass. In the case of xenon, for example, which is the heaviest among the gases under consideration, this number comes to 0.667 at 0.1 MPa, to 0.70 at 2.0 MPa and to 0.866 at 7.0 MPa.

For a binary mixture of inert gases, the Prandtl number is always lower than for its components and varies widely depending on its composition or molecular mass. As molecular mass increases, the Prandtl number drops from the value for the lighter component to a minimum and then shows a steep increase to the value for the heavier component.

The Prandtl number minimum decreases with an increase in the molecular mass of the heavier component and/or with a decrease in the molecular mass of the lighter component of the binary mixture. Thus, the lowest Prandtl number is shown by the He-Xe mixture, with the corresponding minimum of 0.21, and is reached at the molecular mass of 50 g/mole (36 mol % Xe and 64 mol % He).

8. The effect of pressure on transport coefficients and thermodynamic properties of inert gases and their binary mixtures

One can see from the graphs presented in (Tournier and El-Glenk, 2008) that the dependence of transport coefficients and thermodynamic properties of binary inert gas mixtures on the proportion of their components, is by far different from a linear pattern, which points to the inapplicability of a purely additive approach. For example, the dynamic viscosity coefficients of helium-xenon and helium-krypton mixtures show a more or less clearly defined peak, to say nothing of the Prandtl number whose minimum values for a mixture are a factor of 3–4 smaller than those for the pure components of the mixture.

The effect of pressure on transport coefficients and thermodynamic properties of inert gases and their binary mixtures grows with increasing molecular mass. It should be noted, however, that the influence of pressure is more pronounced for pure gases than it is for a binary mixture of the same molecular mass.

For example, at the temperature of 400 K and pressure of 2.0 MPa, the difference between the values of molar heat capacity, specific heat ratio and thermal conductivity coefficient for pure Ar (M = 40 g/mole) and those of the same properties for perfect diluted gas, is equal, respectively, to 2.3%, 1.8%, and 4%, while for the mixture of He_{72}-Xe_{28} (M = 40 g/mole) this difference is under 1%.

9. Algorithms of calculations by semi-empirical formulae for helium-xenon mixtures and their comparison with the Engineering analysis procedure with a fixed mixture composition

The semi-empirical formulae presented in study (Tournier and El-Glenk, 2008) appear in a form unsuitable as yet for practical calculation of transport coefficients and thermodynamic properties of helium-xenon mixtures. In other words, it is still necessary to develop computation algorithms with indication of the calculation sequences, explanation/interpretation of the expressions for the first and second derivatives of the associated model parameters, etc.

The authors of the present paper developed and tested such algorithms on a model of corresponding states of dense gas (CSDG), proceeding from the above semi-empirical relations. In particular, the density calculation algorithm relies on interpretation of expression (6) obtained by the authors through solution of an inverted equation of state (5). The calculation algorithms for dynamic viscosity coefficient, heat conductivity coefficient, and specific heat capacity at constant pressure or constant volume were developed on the basis of expressions (7), (10), (16) and (18), respectively. More detailed information on this subject can be found in (Dragunov et al., 2012).

The developed CSDG algorithms make it possible to calculate the transport coefficients and thermodynamic properties of helium-xenon mixtures regardless of the proportion of components in a mixture. The results of calculations by these algorithms were compared with those obtained by the Simplified Prediction of Thermophysical Properties (SPTP), assuming the same properties of the mixture with a fixed proportion of its components. In accordance with the SPTP, mixture properties are calculated by simple formulae appearing as polynomials.

The correlation was made for the pressure range of 0.1 to 5 MPa and temperature range of 300 to 1700 K, covering the range of helium-xenon mixture parameters of practical interest for various applications. As compared with the CSDG algorithms, the engineering procedure overestimates density and dynamic viscosity coefficient, but does so by not more than 4%. Calculations of heat conductivity coefficient and specific heat capacity at constant pressure were accompanied by systematic deviations in opposite direction, but the understatement of the first and second quantities was no greater than 8% and 2%, respectively.

10. Conclusion

The effort described in (Tournier and El-Glenk, 2008) demonstrated a successful combination of approaches pertaining to the kinetic Chapman-Enskog theory for rarefied gases and the theory of corresponding states for dense gases. Using an extensive experimental database, which covers practically all the known studies, its authors derived empirical expressions for virial coefficients in the equation of state of pure inert gases and their binary mixtures.

At temperatures in excess of (T_{cr} + 40 K), pressures of less than 20 MPa and densities below the critical level, the obtained equation of

state yields values in agreement within 1% with the available experimental data and the NIST recommendations. As would be expected, the accuracy of this equation deteriorates as the critical point becomes closer, which limits its application to temperatures upwards of the critical level.

This equation of state can be used for calculating with the help of fundamental relations various thermodynamic properties of inert gases and their mixtures, such as enthalpy, specific heat capacities C_p and C_v, and specific heat ratio. Unfortunately, no experimental data on these properties have been discovered to date.

Transport coefficients (viscosity and heat conductivity) are found using the two-parameter law of corresponding states, where correction for the density (pressure) effect is an empirical function of the reduced gas density obtained through integration of experimental data by the least-squares method.

For thermal analysis of helium-xenon mixtures, the authors developed CSDG algorithms based on the theory of corresponding states of dense gas and the empirical relations obtained in (Tournier and El-Glenk, 2008)with the use of an extensive array of experimental data.

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